

Samet Sağ

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Mechanical and Materials Engineering graduate from Queen's University, now working in process and quality at Toromont CAT. Experience designing tooling, investigating component failures, and improving shop-floor processes.

EDUCATION

Queen's University

Sep. 2020 – May 2025

BASc, Mechanical and Materials Engineering

Kingston, ON

- GPA: 3.5; Dean's Scholar distinction (twice); Queen's University Excellence Scholarship.

PROFESSIONAL EXPERIENCE

Process & Quality Specialist

Feb. 2026 – Present

Toromont CAT

Bradford, ON

- Investigate failed components and quality issues in heavy-equipment remanufacturing. Identify root causes and put corrective actions in place.
- Design tooling and fixtures for production. Write and update SOPs, work instructions, and quality records to meet ISO 9001 requirements.
- Audit production processes and identify changes to improve product quality.
- Work with production, sales, and engineering staff on manufacturing problems, contamination control, and health and safety.

SparQ Studios Makerspace Supervisor

Feb. 2025 – May 2025

Dunin-Deshpande Queen's Innovation Centre

Kingston, ON

- Supervised the makerspace and helped students use its tools safely.
- Helped users with 3D printing, scanning, woodworking, and metalworking equipment.
- Troubleshoot equipment problems and worked with colleagues on safety procedures and workshop organization.

Mechanical Engineering Intern

May 2023 – Aug. 2024

Toromont CAT

Concord, ON

- Designed stands, fixtures, and components in Inventor. When drawings were unavailable, measured the parts and reverse-engineered their interfaces.
- Designed and implemented an oil dispensing station that cut daily dispensing labour time by 80%. Managed up to 10 engineering and process projects at a time.
- Wrote 10 SOPs and updated more than 40 documents covering manufacturing processes.
- Worked with production staff and outside manufacturers to build tooling and make equipment and workspace changes.
- Mapped shop-floor workflows to find bottlenecks, revise layouts, and write work instructions.

DESIGN TEAM & LEADERSHIP

Chassis/Ergonomics & Brake System Team Member

2021–2023; 2024–2025

Queen's Baja SAE

Kingston, ON

- Led brake design for the 2025 vehicle and mentored two junior members. Earlier chassis work included designing and fabricating a detachable bumper, fuel tank mount, and splash guard.
- Used SolidWorks for design and FEA, and Mastercam/CAM for fabrication.

Industry Relations Team Member

Sep. 2024 – May 2025

Engineering Society of Queen's University

Kingston, ON

- Organized networking and professional development events. Matched students preparing for QUIP internships with past participants through a buddy program.

Ontario Engineering Competition | Consulting

- Represented Queen's University in the consulting category.

ENGINEERING PROJECTS

Weather-Proof Shipping Seals | *Toromont CAT*

- Designed and 3D printed weather seals to replace purchased OEM parts.
- Modeled several seal variants in CAD and set up a stock of printed seals for in-house use.
- Cut unit costs by up to 92% while maintaining protection during shipping.

Component Shipment Stands | *Toromont CAT*

- Designed shipping stands for engines, transmissions, and hydraulic components in Inventor, with drawings for fabrication.
- Worked with vendors to fabricate and deliver more than 10 stands. Maintained over 100 technical drawings.

GPS Tracking of Components | *Toromont CAT*

- Tested satellite and cellular GPS systems and prepared an ROI study to obtain management approval.
- Rolled out hundreds of GPS units across four Eastern Canadian branches to track components in transit.

Component Shipment Photography | *Toromont CAT*

- Built a Python web application prototype with a GUI that captures and saves photos of outbound components from four angles.
- Worked with IT to finish, test, and deploy the application. It replaced manual photo-taking with a consistent shipment record.

Trailer Hitch Finite Element Analysis | *MECH 465*

- Built the hitch geometry in ANSYS APDL and set up loads, boundary conditions, and design constraints.
- Adjusted about 25 variables over successive analyses. The final model had 22% lower maximum stress and 5% less volume.

Rehabilitation Robot End-Effectors | *MECH 460/462 Capstone*

- Designed cable-driven end-effectors and a modular attachment mechanism for a rehabilitation robot.
- Checked the design using FEA in Fusion 360 and revised the geometry.
- Built and integrated the prototype so the robot could use interchangeable end-effectors.

TECHNICAL SKILLS

CAD & Analysis: Autodesk Inventor, SolidWorks, Fusion 360, ANSYS/APDL, finite element analysis

Manufacturing: Mastercam/CAM, GD&T, bills of materials, reverse engineering, rapid prototyping, 3D printing

Quality & Process: Root cause analysis, failure analysis, continuous improvement, process mapping, SOPs, work instructions, ISO 9001 documentation

Programming: Python, MATLAB; experience with C/C++, Java, HTML, CSS

Electrical Prototyping: Basic circuitry, soldering, wiring, breadboards, Wi-Fi boards

ADDITIONAL EXPERIENCE

Sales Representative

TELUS

May – Aug. 2022

Greater Toronto Area

- Spoke with customers at booths and door-to-door about mobile plans and smart home security. Answered questions and processed orders.

Youth Intern

YMCA of Greater Toronto

Jul. – Aug. 2019

Toronto, ON

- Helped program advisors with administration and running sessions for the Newcomer Youth Leadership Development Program.

CERTIFICATIONS & LANGUAGES

Training: Student Leader in Experiential Learning; Introduction to Geometric Dimensioning and Tolerancing; Reverse Engineering Foundations: Product Design; Online Adult, Child, Infant CPR, AED (Level C)

Languages: English and Turkish (native or bilingual proficiency)